

ERIK AMÉZQUITA

Topological Data Analysis (TDA) and shape quantification meet plant biology



CAREER AND EDUCATION

present
|
2025



Assistant Professor

University of Missouri

📍 Columbia, MO

- Division of Plant Science & Technology (100%)
- Department of Mathematics (0%)

2025
|
2023



Preparing Future Faculty Postdoctoral Fellow

University of Missouri

📍 Columbia, MO

- Division of Plant Science & Technology (80%)
- Department of Mathematics (20%)

2023
|
2018



PhD, Computational Mathematics, Science & Engineering

Michigan State University

📍 East Lansing, MI

- Advisors: [Elizabeth Munch](#) and [Dan Chitwood](#)
- Defended: March 2023
- Dissertation: Exploring the Mathematical Shape of Plants

2018
|
2013



Lic. Mathematics (B.S.)

Universidad de Guanajuato

📍 Guanajuato, Gto.

- Advisor: [Antonio Rieser](#) (CONACYT-CIMAT)
- Defended: May 2018
- Thesis: Efficient Object Classification using the Euler Characteristic



PEER-REVIEWED WORK

2026



The Differential Subcellular Localization of Soybean Transcripts, an Additional Regulatory Mechanism of Gene Activity

S. Tennant, E.J. Amézquita, Y. Li, B. Smith, S.S.M.V. Subramanya, S.A. Cervantes-Pérez, S. Thibivilliers, S. Bhattacharya, J. Klaver, M. Libault

- *Genome Biology* 27(9). DOI: [10.1186/s13059-025-03920-z](#)

2024



Decoding the coiling patterns of *Cuscuta campestris* with automated image processing

M. Bentelsbacher, E.J. Amézquita, S. Adhikari, J. Barros, S.Y. Park

- *Plant Cell Reports* 24(282). DOI: [10.1007/s00299-024-03337-1](#)

2024



Allometry and volumes in a nutshell: Analyzing walnut morphology using three-dimensional X-ray computed tomography

E.J. Amézquita, M.Y. Quigley, P.J. Brown, E. Munch, D.H. Chitwood

- *The Plant Phenome Journal* 7: e20095. DOI: [10.1002/ppj2.20095](#)

2023



Genomics data analysis via spectral shape and topology

E.J. Amézquita, F. Nasrin, K.M. Storey, M. Yoshizawa

- *PLoS ONE* 18(4): 30284820. DOI: [10.1371/journal.pone.0284820](#)



View this CV online at
[ejamezquita.github.io/cv](#)

CONTACT

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🔗 [ejamezquita.github.io/](#)

🐙 [ejamezquita](#)

in [erik-amezquita](#)

SKILLS

Programming: Python, R, C/C++, bash/unix

Technologies: *L^AT_EX*, RMarkdown, jupyter, vim, html/css

Languages: Spanish (native), English (fluent), French (elementary)

Made with the R package
[pagedown](#).

Last updated on 2026-05-04.

- 2023 ● **A critical analysis of plant science literature reveals ongoing inequities**
R.A. Marks, E.J. Amézquita, S. Percival, A. Rougon-Cardoso, C. Chibici-Revneanu, S.M. Tebele, J.M. Farrant, R. VanBuren, D.H. Chitwood
• *PNAS* 120(10): e2217564120. DOI: [10.1073/pnas.2217564120](https://doi.org/10.1073/pnas.2217564120)
- 2023 ● **The shape of aroma: measuring and modeling citrus oil gland distribution**
E.J. Amézquita, M.Y. Quigley, T. Ophelders, D. Seymour, E. Munch, D. H. Chitwood
• *Plants, People, Planet* 5(5): 698-711. DOI: [10.1002/ppp3.10333](https://doi.org/10.1002/ppp3.10333)
- 2022 ● **Teaching Tools in Plant Biology. Plants and Python, Coding from Scratch in the Plant Sciences**
R. VanBuren, A. Rougon-Cardoso, E.J. Amézquita, E. Coss-Navarrete, A. Espinosa-Jaime, O. Gonzalez-Iturbe, A. Luckie-Duque, E. Mendoza-Galindo, J. Pardo, G. Rodríguez-Guerrero, P. Rosiles-Loeza, M. Vásquez-Cruz, S. Fernandez-Valverde, T. Hernandez-Hernandez, S. Palande, and D.H. Chitwood
• *The Plant Cell* 34(7): e1. DOI: [10.1093/plcell/koac187](https://doi.org/10.1093/plcell/koac187)
- 2021 ● **Measuring hidden phenotype: Quantifying the shape of barley seeds using the Euler Characteristic Transform**
E.J. Amézquita, M.Y. Quigley, T. Ophelders, J.B. Landis, D. Koenig, E. Munch, D. H. Chitwood
• *in Silico Plants* 4(1): diab033. DOI: [10.1093/insilicoplants/diab033](https://doi.org/10.1093/insilicoplants/diab033)
- 2020 ● **The shape of things to come: Topological data analysis and biology, from molecules to organisms**
E.J. Amézquita, M.Y. Quigley, T. Ophelders, E. Munch, D.H. Chitwood
• *Developmental Dynamics* 249(7): 816-833. DOI: [10.1002/dvdy.175](https://doi.org/10.1002/dvdy.175)



WORK SUBMITTED FOR PEER-REVIEW

- 2025 ● **Tabula Glycine: The whole-soybean single-cell resolution transcriptome atlas**
S.A. Cervantes-Pérez, S. Thibivilliers, S. Amini, J.M. Pelletier, I. Meyer, H. Xu, S. Tennant, P. Ma, C. Sprueil, A.D. Farmer, J.E. Coate, H. Nelissen, Q. Yao, O. Martin, E.J. Amézquita, R.B. Goldberg, J.J. Harada, M. Libault
- 2024 ● **From hand measurements to high throughput phenotyping: understanding maize canopy structure and predicting yield**
Z. Ji, E.J. Amézquita, L. Newton, D.H. Chitwood, A.M. Thompson



RESEARCH STUDENTS MENTORED

- present | 2025 ● **Olivia Fisk and Gloria Asare (Undergraduate)**
Mapper to model rice RNAseq data. Div. of Plant Science. Univ. of Missouri
- present | 2025 ● **Tyler Mitts and Colin Nichols (Undergraduate)**
TDA to model spatial cell distributions. Dept. of Mathematics. Univ. of Missouri
- present | 2025 ● **Sophia Knehans and Roberto Herrera Martin (Undergraduate)**
Mathematical network analysis of academic collaboration. Dept. of Mathematics. Univ. of Missouri.
- 2024 ● **Searcy Thomas and Jake Parmentier (Undergraduate)**
TDA to model spatial cell distributions. Dept. of Mathematics. Univ. of Missouri
- 2024 ● **Ethan Lenhardt (Undergraduate)**
Mathematical network analysis of academic collaboration. Dept. of Mathematics. Univ. of Missouri.



TEACHING EXPERIENCE

- present | 2025
 - **At University of Missouri (as Instructor)**
PLNT_SCI 8001: Special Topics (Computational Modeling for Life Sciences I). Fall 2026.
PLNT_SCI 3550: Data Science for Life Sciences II. Fall 2026.
PLNT_SCI 2500: Data Science for Life Sciences I. Spring 2026.
PLNT_SCI 2002: Special Topics (Data Science for Life Sciences I). Spring 2025
- 2021
 - **At other institutions (as TA)**
[Code In Place](#). Stanford University. Remote. Summer 2021
- 2019
 - **At Michigan State University (as TA)**
[CMSE 201](#): Computational Modelling and Data Analysis I. Fall 2019
- 2016 | 2018
 - **At CIMAT/Universidad de Guanajuato (as TA)**
Precalculus and analytic geometry. Spring 2018
Topology I (Intro to point-set topology). Fall 2017
14th Calculus Problem-solving Workshop. Summer 2017
Introduction to C/C++ and data structures (Online). Summer 2017
Introduction to probability. Fall 2016



INVITED TALKS

- 2026
 - **Exploring the topological shape of plant data**
[SIAM LS26](#). Life Sciences. Cleveland, Ohio
- 2026
 - **The data behind mathematical shapes and plants**
[2026 MUIDSI Symposium](#). MU Institute for Data Science and Informatics. Columbia, MO
- 2026
 - **Explorando la morfología matemática de las plantas**
Seminario Conjunto: Óptica & Física Aplicada. CICESE, Baja California, México
- 2025
 - **Topological Data Analysis to understand Plant Sciences**
[TDA @ MU](#). Department of Mathematics, University of Missouri, Columbia, MO
- 2025
 - **Journeys into interdisciplinarity: How CMSE shaped my career**
[CMSE 10th Anniv.](#). Dept. of Comp. Math, Sci., Eng. Michigan State University, East Lansing, MI
- 2025
 - **Characterizing single-cell transcriptomic spatial patterns with TDA**
[SIAM-AG25](#). Applied Algebraic Geometry. University of Wisconsin, Madison, WI
- 2025
 - **The mathematical shape of plants**
[AATRN](#). Applied Algebraic Topology Research Network. Virtual
- 2025
 - **Mathematically phenotyping shapes and patterns, from molecules to organisms**
[TDA Seminar](#). Dept. of Comp. Math, Science, and Eng. Michigan State University, East Lansing, MI
- 2025
 - **Mathematically phenotyping shapes and patterns, from molecules to organisms**
Plant Science Seminar. Division of Plant Science and Technology. University of Missouri, Columbia
- 2025
 - **The topology of sub-cellular RNA distribution**
[Math & Data Seminar](#). Department of Mathematics. University of Missouri. Columbia, MO
- 2024
 - **The topology of sub-cellular RNA distribution**
[SIAM-MDS24](#). Mathematics of Data Science. Atlanta, GA

- 2024 ● **Characterizing spatial patterns and distributions with Topological Data Analysis (TDA)**
[NAPPN AI/ML Affinity Group](#). North American Plant Phenotyping Network. Virtual
- 2023 ● **Mapper and the topological shape of genomic analysis**
[MU-GNU International Symposium](#) in Plant Biotechnology. Bond LSC. Columbia, MO
- 2023 ● **A primer on Topological Data Analysis**
Geometry and Topology Seminar. Department of Mathematics. University of Missouri. Columbia, MO
- 2023 ● **Exploring the mathematical shape of plants**
[CS Colloquium](#). Department of Computer Science. Saint Louis University. St. Louis, MO
- 2023 ● **When topology meets plant morphology**
[USTARS 2023](#). Underrepresented Students in Topology and Algebra Research Symposium, Seattle
- 2023 ● **The mathematical shape of plants**
Plant Sciences Seminar. Department of Botany and Plant Sciences. University of California, Riverside
- 2023 ● **Directional statistics to describe the distribution of citrus oil glands**
[JMM 2023](#). Joint Mathematics Meeting. American Mathematical Society. Boston, MA.
- 2022 ● **The mathematical shape of plants**
Plant Science Seminar. Division of Plant Science and Technology. University of Missouri, Columbia
- 2022 ● **Using applied topology in plant science**
[Stochastic Topology seminar](#). Max Planck Institute for Mathematics in the Sciences (MiS). Virtual.
- 2022 ● **TDA to harness plant morphology**
[Multicellular dynamics seminar](#). Max Planck Institute for Plant Breeding Research (MIPZ). Virtual
- 2022 ● **Using the Euler characteristic to quantify the shape of barley seeds**
[OU Topology and Data Science Seminar](#). Department of Math. University of Oklahoma. Virtual
- 2022 ● **Bridging applied topology and plant biology**
[JMM 2022](#). Joint Mathematics Meeting. American Mathematical Society. Virtual
- 2022 ● **Measuring the shape of plants with the Euler Characteristic Transform**
[UFTDA 2022](#). University of Florida Topological Data Analysis Conference. Gainesville, FL
- 2021 ● **Analyzing maize leaf angles and modeling leaf curvature**
[2021 NAPPN](#). North American Plant Phenotyping Network. Virtual
- 2018 ● **TDA to bridge mathematics and archaeology**
[16th School on Probability and Statistics](#). CIMAT. Guanajuato. Gto.
- 2017 ● **Efficient object classification using the Euler characteristic**
[II Coloquio](#) de Desarrollo Tecnológico al Servicio del Patrimonio Cultural. Guanajuato. Gto.



RECENT CONTRIBUTED TALKS

- 2025 ● **Modeling differential sub-cellular localization of transcripts**
[Plant Biology 2025](#). American Society of Plant Biologists. Milwaukee, WI

- 2025 ● **Characterizing single-cell transcriptomic spatial patterns with TDA**
[MW-ASPB 2025](#). ASPB Midwest Section. University of Nebraska, Lincoln, NE
- 2024 ● **Topological Data Analysis to model spatial data**
[MMBS](#). Midwest Mathematical Biology Seminar. Virtual.
- 2024 ● **Topological Data Analysis to characterize transcriptomic spatial distributions**
[SIAM-CSS24](#). SIAM Central States Section. University of Missouri, Kansas City, MO
- 2024 ● **The early dodder gets the host**
[MW-ASPB 2024](#). ASPB Midwest Section. Purdue University. West Lafayette, IN
- 2023 ● **The wal(nut)zing nutcracker: linking morphological and commercial traits in walnuts**
[IPG Plant Talks](#). Interdisciplinary Plant Group. University of Missouri. Columbia, MO



WORKSHOPS LEAD

- 2022 ● **The shape of things: Measuring the shape of plants with Topological Data Analysis**
[2022 NAPPN](#). North American Plant Phenotyping Network. Athens, GA. [Check material](#).
- 2021 ● **Using the Euler characteristic to quantify the shape in biology**
[2021 AATRN Tutorial-a-thon](#). Applied Algebraic Topology Research Network. [Watch video](#).
- 2021 ● **Measuring the shape of plants with Topological Data Analysis**
[2021 NAPPN](#). North American Plant Phenotyping Network. [Check material](#).



SELECT FLASH TALKS (WITH POSTERS)

- 2024 ● **Topological Data Analysis to characterize transcriptomic spatial distributions**
[CAFNR Research Symposium](#). College of Ag., Food, and Nat. Resources. University of Missouri
- 2024 ● **The shape and size of shells, kernels, and cracks, in a nutshell**
[NAPPN 2024](#). North American Plant Phenotyping Network. West Lafayette, IN
- 2023 ● **Modeling the shape of citrus and their oil gland distribution**
[NAPPN 2023](#). North American Plant Phenotyping Network. St. Louis, MO
- 2022 ● **Using topology to analyze the shape of plants**
[IPPS2022](#). International Plant Phenotyping Symposium. Wageningen, The Netherlands
- 2020 ● **Quantifying barley morphology**
[TDA@NeurIPS](#). Neural Information Processing Systems. Online.
- 2020 ● **Using Euler Characteristic Curves to model barley shape**
[YRF@SoCG](#). Young Researcher Forum @ CG Week, Symposium on Computational Geometry. Virtual



SELECT WORKSHOPS AND HACKATHONS ATTENDED

- 2025 ● **Building Bridges to Advance Data Science Education**
[4th Annual BEDE Network Meeting](#). University of California, Santa Barbara, CA
- 2023 ● **Graduate Wellness and Mental Health Ambassador Program.**
The Graduate School. Michigan State University, East Lansing, MI

- 2022 ● **NatSci Cultural Competency. Workshop Semester Series**
DEI Office. College of Natural Science. Michigan State University. East Lansing, MI
- 2022 ● **Beyond Abstract Measures: geometry and computation**
Organized by the [Lorentz Center](#), Leiden, The Netherlands
- 2021 ● **Datathon4Justice**
[D4J@QSIDE](#). Institute for Quantitative Study of Inclusion, Diversity, and Equity. Virtual
- 2021 ● **Immersive Visualization Institute**
[IVI2021](#). Abrams Planetarium, MSU Libraries, and MSU Museum. East Lansing. MI
- 2021 ● **MSU Dialogues: Race. Semester I**
[Office for Institutional Diversity and Inclusion](#). Michigan State University. Virtual
- 2019 ● **Applied Mathematical Modeling with Topological Techniques.**
[ICERM](#). Institute for Computational and Experimental Research in Mathematics. Providence, RI

+ OUTREACH

- 2023 ● **If life gives you lemons, analyze the shape of their aroma**
Science on Tap. International Tap House. Columbia, MO
- 2023 ● **Un matemático y un botánico van por una limonada**
¡Science on Wheels en Español! SACNAS Mizzou. Columbia, MO
- 2023 ● **Mental Health in Mathematics and Computer Science**
Panel organizer and moderator. [SGI23](#). Massachusetts Institute of Technology. Virtual
- 2022 ● **Webinar de Solicitudes al Doctorado en Estados Unidos**
Panelist. Organized by the Coloquio de Exestudiantes CIMAT/DEMAT. Virtual
- 2022 ● **Mental Health in Mathematics and Computer Science**
Panel organizer and moderator. [SGI22](#). Massachusetts Institute of Technology. Virtual
- 2021 ● **A topologist and a plant biologist go for a newly shaped beer**
[Hispanics in STEM celebration](#). WaMPS. Michigan State University. East Lansing, MI
- 2021 ● **Demeter y Euler van por una cerveza**
[Seminario Junior de Estudiantes](#). Departamento de Matemáticas. Universidad de Guanajuato. Virtual.
- 2020 ● **Cuantificando la forma de la cebada con ATD**
Seminario de Matemáticas y Estadística. Instituto Politécnico Nacional. Virtual
- 2020 ● **Using topology to quantify the shape of barley**
[Summer Math Academy](#). Math Department. University of Toronto. Virtual
- 2020 ● **Wrangling and Presenting Data with Pandas and Seaborn in Python**
[Social Science Data Analytics Initiative](#). Michigan State University. Virtual
- 2020 ● **Narrating our data with RMarkdown**
[Social Science Data Analytics Initiative](#). Michigan State University. Virtual

- 2018 ● **La maldición de la dimensión y aprendizaje de máquina**
Ciencia es Cultura. Dirección de Extensión Cultural. UGto. San Luis, Gto.
- 2017 ● **Un matemático y un psicólogo se hallan en Hanoi**
Ciencia es Cultura. Dirección de Extensión Cultural. UGto. Guanajuato, Gto.
- 2016 ● **Infinitos grandes e infinitos pequeños**
Ciencia es Cultura. Dirección de Extensión Cultural. UGto. San Miguel Allende, Gto.



SELECT SERVICE

- present
|
2024 ● **In-house data scientist**
Division of Plant Science and Technology. University of Missouri
• Management and rearrangement of various databases concerning several projects carried out by DPST.
- present
|
2024 ● **Faculty Search Committee (Data Science Instructor)**
Division of Plant Science and Technology. University of Missouri
- 2025
|
2024 ● **MU Plant Research Symposium Organizing Committee**
Webmaster for the [9th Annual MU-Corteva Plant Research Symposium](#). University of Missouri
- 2023 ● **Mentor for [BIPS](#): Bioinformatics in Plant Science**
Division of Plant Science and Technology. University of Missouri
• Lead weekly workshops on good coding practices. Geared toward plant science undergraduate students.
- 2023
|
2021 ● **Mentor for [SGI](#) REU: Summer Geometry Initiative**
Geometric Data Processing Group. Massachusetts Institute of Technology. Remote
• General technical support. Organized panels on Mental Health for the 2022 and 2023 editions.
- 2023
|
2022 ● **CMSE Chair Search Committee**
Department of Computational Mathematics, Science, and Engineering. Michigan State University.
- 2022 ● **Mentor for the [ACRES](#) REU: Advanced Computational Research Experience**
Institute for Cyber-enabled Research. Michigan State University.
• Conducted weekly professional development workshops, such as CV and academic poster design.
- 2022
|
2021 ● **President of the CMSE Graduate Student Organization**
Council of Graduate Students. Michigan State University
• Lead department-wide events, committees, and inquiries to attend graduate students' needs
- 2017
|
2016 ● **Student Representative**
College of Natural and Exact Sciences Council. Universidad de Guanajuato.
• Logged each session minutes, such as budget or policy, and shared them with the math students.
- 2016
|
2015 ● **High School Mathematics Seminar Co-Organizer**
Escuela de Nivel Medio Superior, Guanajuato. Guanajuato.
• Organized lectures on college-level math topics, such as combinatorics or group theory.



SELECT AND RECENT AWARDS

- 2025 ● **Travel Grant (US\$600)**
[BEDE Network Annual Meeting](#). Biological and Environmental Data Education. Santa Barbara, CA
- 2025 ● **Travel Grant (US\$575)**
[Plant Biology 2025](#). American Society of Plant Biologists. Milwaukee, WI
- 2024 ● **Travel Grant (US\$250)**
[SIAM-CSS24](#). SIAM Central States Section. Kansas City, MO.
- 2024 ● **Travel Grant (US\$650)**
[SIAM-MDS24](#). Early Career Award. SIAM-Mathematics of Data Science. Atlanta, GA
- 2024 ● **Travel Grant (US\$350)**
[MW-ASPB 2024](#). ASPB Midwest Section. West Lafayette, IN
- 2024 ● **Best Flash Talk (1st place out of 52 talks)**
[2024 NAPPN](#). North American Plant Phenotyping Network. West Lafayette, IN
- 2023 ● **Distinguished Graduate Student. Travel Grant (US\$700)**
[USTARS 2023](#). Underrepresented Students in Topology and Algebra Research Symposium.
- 2022 ● **Best Poster Award. 3rd place out of 173 posters.**
[IPPS2022](#). International Plant Phenotyping Symposium. Wageningen, The Netherlands
- 2022 ● **Travel Grant (EUR 2000)**
[IPPS2022](#). International Plant Phenotyping Symposium. Wageningen, The Netherlands
- 2022 ● **Fitch H. Beach Award (2nd place)**
College of Engineering. Michigan State University. Most outstanding graduate research.
- 2022 ● **Travel Grant (US\$800)**
[2022 NAPPN](#). North American Plant Phenotyping Network. Athens, GA
- 2019 ● **Travel Grant (US\$800)**
[Applied Mathematical Modeling with Topological Techniques](#). ICERM. Providence, RI
- 2019 ● **IMPACTS Fellowship**
Awarded jointly by Michigan State University and the NRT-NSF program (NSF DGE-1828149).
- 2018 ● **Sotero Prieto Medal**
Sociedad Mexicana de Matemáticas. Best undergrad math thesis produced in Mexico
- 2018 ● **Francisco Aranda Ordaz Award (3rd place)**
Asociación Mexicana de Estadística. Best undergrad statistics thesis produced in Mexico.
- 2018 ● **Raymond P. and Marie M. Ginther Graduate Fellowship**
Awarded by CMSE to outstanding incoming graduate students.
- 2018
|
2013 ● **CIMAT Academic Excellence Scholarship**
Merit-based scholarship for math undergraduates.

2017

- **Best Undergraduate Mathematics, Physics and Earth Sciences Research Project (3rd place).**
4to Congreso Interinstitucional de Jóvenes Investigadores.

2017

- **Best Undergraduate Engineering Research Project (1st place)**
5to Encuentro de Jóvenes Investigadores. Universidad de Guanajuato.